INCH-POUND

MIL-M-48185B(AR)
25 September 1992
SUPERSEDING
MIL-M-48185A(PA)
30 March 1976

MILITARY SPECIFICATION

MUNITION FOR PROJECTILE, 155MM, HE, M692/M731, DETONATING CORD ASSEMBLY FOR

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not normally covered by the drawings, and provides quality assurance provisions for the fabrication of parts, loading, assembling and packing for one type of Detonating Cord Assembly used in the munition for Projectiles, 155MM, M692 and M731. (See 6.1)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (See 6.2)

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1375

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STANDARDS

MILITARY

MIL-STD-453 - Inspection, Radiographic
MIL-STD-650 - Explosive: Sampling, Inspection and Testing
MIL-STD-410 - Non-Destructive Testing Personnel Qualification and Certification (Eddy Current, Liquid Penetrant Magnetic Particle, Radiographic and Ultrasonic.)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in cited in the solicitation.

DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9245691 - Packing and Marking for Box, Inner
9324713 - Packing and Marking for Box, Outer
9324715 - Packing and Marking for Box, Packing
9245691 - Electric Detonator, XM100
9298406 - Detonating Cord Assembly

PUBLICATIONS

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-199

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with the applicable drawings and specifications.

3.2 Detonating cord assembly. The detonating cord assembly shall comply with all requirements specified on dwg. 9298406, all associated drawings, and with all requirements specified in applicable specification.

3.3 Radiographic examination of detonating cord. The detonating cord shall not contain any voids, inclusions, gaps, or impurities in the explosive core or metal sheath when radiographically examined.

3.4 Impact resistance of detonating cord. The detonating cord when placed on a steel plate and impacted with a twenty (20) pound steel ball dropped from a height of thirty (30) inches shall not detonate or deflagrate.

3.5 Functioning of detonating cord assembly. The detonating cord assembly shall function and propagate uniformly along its entire length.

3.6 High temperature conditioning and functioning of detonating cord assembly. After high temperature conditioning at 165 + 5°F, the detoning cord assembly shall function and propagate uniformly along its entire length while at a temperature of +125 ± 5F.

3.7 Low temperature conditioning and functioning of detonating cord assembly. When low temperature conditioned at -25 + 5F, the detonating cord assembly shall function and propagate uniformly along its entire length.

3.8 Use of adhesives and sealants. The contractor shall develop a documented operational procedure for utilization of adhesives and sealants. When applicable, this procedure shall provide for quality control of raw materials and process to be used; details associated with intended use, such as proper handling, mixing, curing cycles, pot life, etc., cleaning agents, safety precautions, special ventilation, drying or other processes, and any other information of a similar nature that is considered necessary for the particular adhesive or sealant in use. This procedure shall be approved by the Government contracting officer prior to use, and whenever a change is made to the approved sequence such that the original assurance of product quality is affected.

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3.9 First article inspection. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.10 Workmanship - detonating cord assembly. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, discoloration and other defects which would make the item unsuitable for its intended use.

3.11 Sheath wall thickness. The sheath wall thickness shall not be less than .0074 inches nor greater than .0159 inches.

4 QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspections, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First Article Inspection. (See 4.3)

b. Quality Conformance Inspection. (See 4.4)

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Detonating cord assembly	9298406	72
Detonating cord	9298409	31 feet

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. See MIL-A-48078. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Detonating cord assembly and components		SHEET 1 OF 1		DRAWING NUMBER
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Detonating cord (dwg. 9298409) 31 feet		Sample length		
	Examination for defects	10	9 in.	3.2	4.4.2.2 Note 1
	Flexibility for detonation velocity	10	9 in.	3.2	4.5.6
	Radiographic examination	10	12 in.	3.3	4.5.4
	Core load Determination	2	6 in.	3.2	4.5.5
	Impact resistance test	10	10 in.	3.4	4.5.11
	Detonating cord assembly (dwg. 9298406)				
	Examination for defects	50		3.2	4.4.2.8
	Functioning test	32	32-0-1	3.5	4.5.8
	High temperature conditioning and functioning test	20	20-0-1	3.6	4.5.9
	Low temperature conditioning and functioning test	20	20-0-1	3.7	4.5.10

NOTES: 1. These samples to be used for flexibility and detonation velocity test.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition inspection lot of mines shall contain:

a. Detonating cord produced in one continuous process.

b. RDX from one batch or blending of batches which results in homogeneity of material used in the detonating cord.

c. Sheathing material of one lot interfix number from one manufacturer.

d. PBXN-5 from one batch or blending of batches which results in homogeneity of material used in the booster cup assembly.

e. Booster cups from one lot interfix number from one manufacturer.

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE II Atttributes sampling inspection

Lot Size	Inspection Level					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13

Numbers under inspection levels indicate sample size; asterisks indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of samplin procedures, etc., may be used by the contractor when they provide as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	RDX	1 OF 1	NA
			NEXT HIGHER ASSEMBLY 9298409

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Moisture and volatiles content		3.2	4.5.2
Minor 201	Poor workmanship	Level IV	3.10.2	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.2	TITLE Cord, Detonating	SHEET 1 OF 1	DRAWING NUMBER 9298409
			NEXT HIGHER ASSEMBLY 9298407

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
101	Diameter	Level II	3.2	Gage
102	Sheath wall thickness	Level II	3.11	Gage
103	Sharp bend or kink in cord	Level II	3.2	Visual
104	Radiographic examination	10, Note 1	3.3	4.5.4
105	Core load	2, Note 2	3.2	4.5.5
106	Flexibility and detonation velocity	10, Note 3	3.2	4.5.6
107	Impact resistance test	10, Note 4	3.4	4.5.11
Minor				
201	Poor workmanship	Level IV	3.10.2	Visual

NOTES:
Note 1: Samples from beginning and end of each continuous length from one fill tube or process load. Length-12 inch.
Note 2: sample length - 6 inches
Note 3: Same as Note 1 except length to be 9 inches.
Note 4: Same as Note 1 except length to be 6 inches.

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-48185B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Ferrule	1 OF 1	9298408
			NEXT HIGHER ASSEMBLY 9298407

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Diameter of hole	Level II	3.2	Gage
Minor				
201	Width of groove	Level IV	3.2	Gage
202	Depth of groove	Level IV	3.2	Gage
203	Finish improper	Level IV	3.2	Visual
204	Presence of burrs or sharp corners	Level IV	3.2	Visual
205	Poor workmanship	Level IV	3.10.2	Visual

NOTES:

AMSMC Form 1570a, 1 Jul 89

Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Ferrule and cord assembly	1 OF 1	9298407
			NEXT HIGHER ASSEMBLY 9298406

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Pull test of stake		3.2	4.5.7
102	Total length	Level II	3.2	Gage
103	Length of ferrule, max.	Level II	3.2	Gage
104	Core explosive broken out or sheath material smeared or closing off core	Level II	3.2	Visual
105	Detonating cord above flush with ferrule	Level II	3.2	Visual/Gage
106	Stake missing or incomplete (2)	Level II	3.2	Visual
107	Crimp missing or improper	Level II	3.2	Visual
108	End of cord not closed	Level II	3.2	Visual
Minor				
201	Core closure plug missing (when applicable)	Level IV	3.2	Visual
202	Poor workmanship	Level IV	3.10.2	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH 4.4.2.5	TITLE Cup, booster	SHEET 1 OF 1	DRAWING NUMBER 9298411
			NEXT HIGHER ASSEMBLY 9298410

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Thickness at closed end	Level II	3.2	Gage
102	Length	Level II	3.2	Gage
Minor				
201	True position of inside diameter with outside diameter	Level IV	3.2	Gage
202	Poor workmanship	Level IV	3.10.2	Visual

NOTES:

AMSMC Form 1570a, 1 Jul 89 Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.6	TITLE PBXN-5	SHEET 1 OF 1	DRAWING NUMBER NA
			NEXT HIGHER ASSEMBLY 9298410

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major 101	Moisture and total volatile content		3.2	4.5.1
Minor 201	Poor workmanship	Level IV	3.10.2	Visual

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CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH 4.4.2.7	TITLE Booster cup assembly	SHEET 1 OF 1	DRAWING NUMBER 9298410
			NEXT HIGHER ASSEMBLY 9298407

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Height of explosive charge	Level II	3.2	Gage
102	Density of explosive charge		3.2	4.5.3
103	Outside diameter	Level II	3.2	Gage
Minor				
201	Cup damaged (deformed or punctured)	Level IV	3.2	Visual
202	Poor workmanship	Level IV	3.10.2	Visual

NOTES:

AMSMC Form 1570a, 1 Jul 89

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.8	TITLE Detonating cord assembly	SHEET 1 OF 1		DRAWING NUMBER 9298406
				NEXT HIGHER ASSEMBLY NA
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Cord sheath cracked	100%	3.2	3 power mag.
102	Voids or surface cracks in epoxy	100%	3.2	3 power mag.
103	Temperature conditioning and functioning test			
	a. Ambient	20	3.5	4.5.8
	b. High temperature	10, Note 1	3.6	4.5.9
	c. Low temperature	10	3.7	4.5.10
104	Cup diameter, max.	Level II	3.2	Visual
105	Booster cup assembly damaged (dented, cracked, or punctured)	Level II	3.2	Visual
106	Contour in cord missing	Level II	3.2	Visual
107	Crimped around booster cup missing or incomplete	Level II	3.2	Visual
Minor				
201	Epoxy incomplete	Level IV	3.2	Visual
202	Straightness of detonating cord	Level IV	3.2	STME
203	Poor workmanship	Level IV	3.10.1	Visual

NOTES: (1) Lot acceptance test plan shall be 40-0-2. If any failure occurs, a second sample of the same quantity at the test environment where failure occurred is permitted. Lot is rejected if two (2) or more failures occur in the total quantity tested.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Box, inner prior to sealing	1 OF 1	9294711
			NEXT HIGHER ASSEMBLY: --

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Incorrect number of detonator cord assemblies	Level II	3.2	Visual
102	Assembly inverted	Level II	3.2	Visual
Minor				
201	Cover missing	Level IV	3.2	Visual
202	Marking incorrect or unidentifiable	Level IV	3.2	Visual
203	Poor workmanship	Level IV	3.10.2	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.10	TITLE Box, outer prior to sealing	SHEET 1 OF 1		DRAWING NUMBER 9294713
				NEXT HIGHER ASSEMBLY --
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Incorrect number of inner boxes	Level II	3.2	Visual
102	Spacer missing	Level II	3.2	Visual
103	Fillers missing	Level II	3.2	Visual
Minor				
201	Poor workmanship	Level IV	3.10.2	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Box, outer, sealed prior to inserting into bag	1 OF 1	MIL-M-48195B(AR) 9324713
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Corners not blunted	Level II	3.2	Visual
102	Sealing strip missing	Level II	3.2	Visual
Minor				
201	Marking incorrect or unidentifiable	Level IV	3.2	Visual
202	Poor workmanship	Level IV	3.10.2	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-M-48185B(AR)

PARAGRAPH 4.4.2.12	TITLE Sealed barrier bag	SHEET 1 OF 1	DRAWING NUMBER 9324713
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Heat seal of barrier bag missing or incomplete	Level II	3.2	Visual
102	Barrier bag torn or punctured	Level II	3.2	Visual
Minor				
201	Marking incorrect or unidentifiable	Level IV	3.2	Visual
202	Contents loose	Level IV	3.2	Visual/Manual
203	Poor workmanship	Level IV	3.10.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Packing and marking for box, Packing prior to sealing	1 OF 1	9324713
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Incorrect number of bagged outer boxes	Level II	3.2	Visual
Minor				
201	Fillers missing	Level IV	3.2	Visual
202	Poor workmanship	Level IV	3.10.2	Visual

NOTES:

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MIL-M-48185B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Box, packing, sealed	1 OF 1	9324715
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Boxed damaged exposinh contents	Level II	3.2	Visual
102	Strapping missing, broken or loose	Level II	3.2	Visual
Minor				
201	Marking incorrect or unidentifiable	Level IV	3.2	Visual
202	Poor workmanship	Level IV	3.10.2	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078.

4.5 Methods of inspection.

4.5.1 Moisture and volatiles content of charge in booster cup assembly. The contractor shall provide adequate controls to insure compliance with the applicable requirements. The moisture and volatiles content of the explosive charge shall be determined at the loading station at the time of loading. This determination shall be made at least once each four (4) hours of production. Sampling of the explosive charge shall be in accordance with MIL-STD-650, Method 101.5. In addition, if the moisture and volatiles content fails to comply with the requirement specified on the applicable drawing, and loading has not begun, the explosive charge material shall be rejected until corrective action has been taken as verified by repeating the moisture and volatiles content determination. If loading has begun, booster cup assemblies loaded with the non-conforming explosive charge material shall be rejected.

4.5.2 Moisture and volatiles content of RDX for detonating cord. The contractor shall provide adequate controls to insure compliance with the applicable requirements. The moisture and volatiles content of the RDX shall be determined at the loading station at the time of loading. This determination shall be made at least once each four (4) hours of production. Sampling of the RDX shall be in accordance with MIL-STD-650, Method 101.5 In addition, if the moisture and volatiles content fails to comply with the requirement specified on the applicable drawing, and loading has not begun, the RDX shall be rejected until corrective action has been taken as verified by repeating the moisture and volatiles content determination. If loading has begun the detonating cord loaded with the non-conforming RDX shall be rejected.

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4.5.3 Density of explosive charge in booster cup assembly. A minimum of two (2) assemblies, representative of the booster cup assemblies being processed, shall be loaded and the densities determined to qualify each press prior to its use in production. In addition, a sample of two (2) assemblies shall be processed during the last portion of each one (1) hour's production from each press and the density determined as a control of each press during production.

If any defective assembly is found when inspection is performed for control of a press, the press producing the defective assembly shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All presses removed from production because of some fault, may, after corrective action, be returned to production providing they pass the qualification test above.

The following procedure shall be used to determine the density of the explosive charge.

a. Weigh empty booster cup (grams)

b. Measure bottom thickness (inches)

c. Measure wall thickness (inches)

d. Load cup

e. Weigh loaded booster cup (grams)

f. Measure height to top of charge (inches)

g. Measure outside diameter after loading (inches)

h. Calculate weight of charge (e-a) (grams)

i. Calculate height of charge (f-b) (inches)

j. Calculate volume of explosive charge material (cubic centimeters) = (.7854) x 16.39) x (i) x (g-2c) x (g-2c)

k. Calculate density of explosive charge material (grams/cubic centimeters) = h/j

WARNING: The following tests are conducted on assemblies containing an explosive component. Test barricade (where used), procedures and equipment shall have prior approval.

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4.5.4 Radiographic examination of detonating cord. Radiographic equipment procedures and evaluation shall be in accordance with MIL-STD-453. Radiographic personnel shall be qualified in accordance with MIL-STD-410.

Samples for this test shall be selected from each end of the continuous length produced from one fill tube or process load. Twelve (12) samples will be selected from each detonating cord lot. Each sample shall be 1.0 feet long. The minimum optical density of each radiograph in the area being inspected shall not be less than 1.25 and shall be within a density range that will permit proper identification of defects. The radiographs shall be examined for the condition specified in 3.3. Any sample with any one of the specified conditions shall be classed defective.

Samples that satisfy requirements may be used in the flexibility and detonation velocity test specified in 4.5.6 or the core load test specified in 4.5.5.

A standard detonating cord shall be fabricated and exposed on each radiographic film. The standard shall consist of a piece of detonating cord three inches long with one-quarter inch of explosive removed from each end. A satisfactory radiographic exposure must clearly show the outline of the inside and outside diameters of the aluminum sheath and both ends of the explosive core inside the sheath.

4.5.5 Core load of detonating cord. After completion of the test specified in 4.5.4, the samples of detonating cord shall be weighed before and after dissolving out the explosive core, using an analytical balance or other suitable weighing equipment. The difference in weight (in grains) divided by the length (in feet) is the core load. Units in which the core load (grains/foot) is outside the limits specified on the applicable drawing shall be classed defective.

4.5.6 Flexibility and detonation velocity of detonating cord. After completion of the test specified in 4.5.4 the samples of detonating cord shall be bent ninety degrees (90°) around a 0.188 inch maximum diameter mandrel and initiated by a number 6 commercial blasting cap, or equivalent, taped to one end of the detonating cord. After bending about the mandrel, the cord shall be examined for any fracture of the sheath. Any cord in which the sheath fractures shall be classed defective.

The detonation velocity shall be measured while the cord is bent about the mandrel using an electronic time intervelometer, oscilloscope, electronic timer, or similar equipment suitable for determining the velocity. The detonation velocity of each sample shall be recorded and any sample failing to comply with the minimum velocity shall be classed as a failure.

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4.5.7 Pull test of stake of ferrule and cord assembly. The sample assemblies for this test shall be selected from each staking station at the start of production each day and after each one (1) hour of production or portion thereof. A sample of a least five (5) ferrule and cord assemblies shall be tested to qualify each tool used in the staking process for use in production. In addition, a random sample of two (2) parts shall be selected from the last portion of each hour's production for testing as a control of the tool during production.

If any defective assemblies are found during qualification of the tool, the tool producing the defective assemblies shall not be used in production.

If any defective assemblies are found when testing is performed for control of the tool, the tool producing the defective assembly shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to t Contracting Officer to substantiate the request.

The samples to be subjected to this test shall be tested by using a fixture to hold the assembly in a vertical position. A 0.5 + 0.05 pound weight shall be suspended from the ferrule. The weight shall be applied as a static load and shall not be moved after attachment of the ferrule. Assemblies in which the ferrule shifts on the cord or in which there is any cracking of the detonating cord at the ferrule interface shall be classed defective. This is a destructive test. Any assembly so tested shall not be returned to the lot.

4.5.8 Functioning test of detonating cord assembly. The detonating cord assembly shall be initiated by the output of an M100 Electric Detonator from an accepted lot. The detonator and booster of the detonating cord assembly shall all be axially aligned in one plane with an air gap of .230 + 0.10 between the detonator and booster. A suitable fixture shall be used to maintain the air gap and the axial alignment. The detonating cord portion of the assembly shall be placed in contact with a 6061-T6 aluminum witness plate having a 0.060 inch minimum thickness. The cord shall be secured to the plate with tape. The test assembly shall be placed behind an approved barricade, and the detonator short shall be removed. The detonator shall be fired using a 6 volt DC source (or other suitable source).

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Any detonating cord assembly which fails to function, fails to propagate uniformly along its entire length, or in which the last one (1) inch of the detonating cord fails to produce a groove in the witness plate shall be classed as a failure.

Units which the detonator fails to function *DUD) shall be classed as a "No Test" and another detonator used to initiate the detonating cord assembly.

4.5.9 High temperature conditioning and functioning test of detonating cord assembly. The assemblies to be subjected to this test shall be placed in a suitable oven and the temperature increased to 165 + 5°F with relative humidity not less than 15%. the rate of temperature increase shall not exceed 18°F per minute. The samples shall be maintained at the conditioning temperature for forty eight (48) hours minimum. After the time has elapsed the temperature shall be reduced to +125 + 5°F and the assemblies maintained at temperature for two (2) hours minimum. The assemblies shall then be removed from the oven and functioned as specified in 4.5.8. Set-up time between removal from the oven and firing shall be five (5) minutes maximum. Any detonating cord assembly which fails to function, fails to propagate uniformly along its entire length, or in which the last one (1) inch of the detonating cord fails to produce a groove in the witness plate shall be classed as a failure.

4.5.10 Low temperature conditioning and functioning test detonating cord assembly. The assemblies to be subjected to this test shall be placed in a suitable chamber and the temperature decreased to -25 + 5°F. The assemblies shall be conditioned for 2 hours minimum, removed from the conditioning chamber and functioned as specified in 4.5.8. Set-up time between removal from the oven and firing shall be five (5) minutes maximum. Any detonating cord assembly which fails to function, fails to propagate uniformly along its entire length, or in which the last one (1) inch of the detonating cord fails to produce a groove in the witness plate shall be classed as a failure.

4.5.11 Impact resistance test of detonating cord. The sample of detonating cord shall be placed between two steel plates (2 inch by 2 inch x 0.25 inch) supported by a solid base and the upper plate impacted by a solid steel ball weighing 20.0 + 0.5 lb dropped from a height of 30.0 + 0.5 inches. If any sample detonates or deflagrates upon impact it shall be classed as a failure.

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5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level C. The detonating cord assemblies shall be preserved and packaged in accordance with dwg. 9324711.

5.2 Packing.

5.2.1 Level C. The detonating cord assemblies preserved and packaged as specified in 5.1.1, shall be packed in accordance with dwgs. 9324711, 9324713, 9324715 and Code of Federal Regulations, Title 49, Parts 100-199.

5.3 Shipping. When assemblies from more than one lot are shipped at one time, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. The components and assembly covered by this specification are intended for use as the power supply for the housing, timer and fuzing for the projectile, 155MM, M692/M731.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced in section 2.1.

6.3 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor to evaluation to the procuring contracting officer.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as, US Army Armament Research and Development Command, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Submission of inspection equipment designs for approval. (See MIL-A-48078). Equipment designs shall be submitted as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAT-I (D), Picatinny Arsenal, New Jersey 07806-5000. Request letter of submittal should state contractor, contract number, specification number, item nomenclature, and classification of defects and tests paragraph number. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.6 Subject term (keyword) listing.

Radiographic
RDX
PBXN-5
Sheath
Ferrule

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian
Army-AR

Preparing activity:
Army-AR

(Project 1375-A207)

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Arsenal, Dover, New Jersey, ATTN: SARPA-FR-S for the Propellant Sur-

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revisi letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-48185	25 SEPTEMBER 1992

3. DOCUMENT TITLE
MUNITION FOR PROJECTILE, 155MM, HE, M692/M731, DETONATING CORD ASSEMBLY FO

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMACA-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/20

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